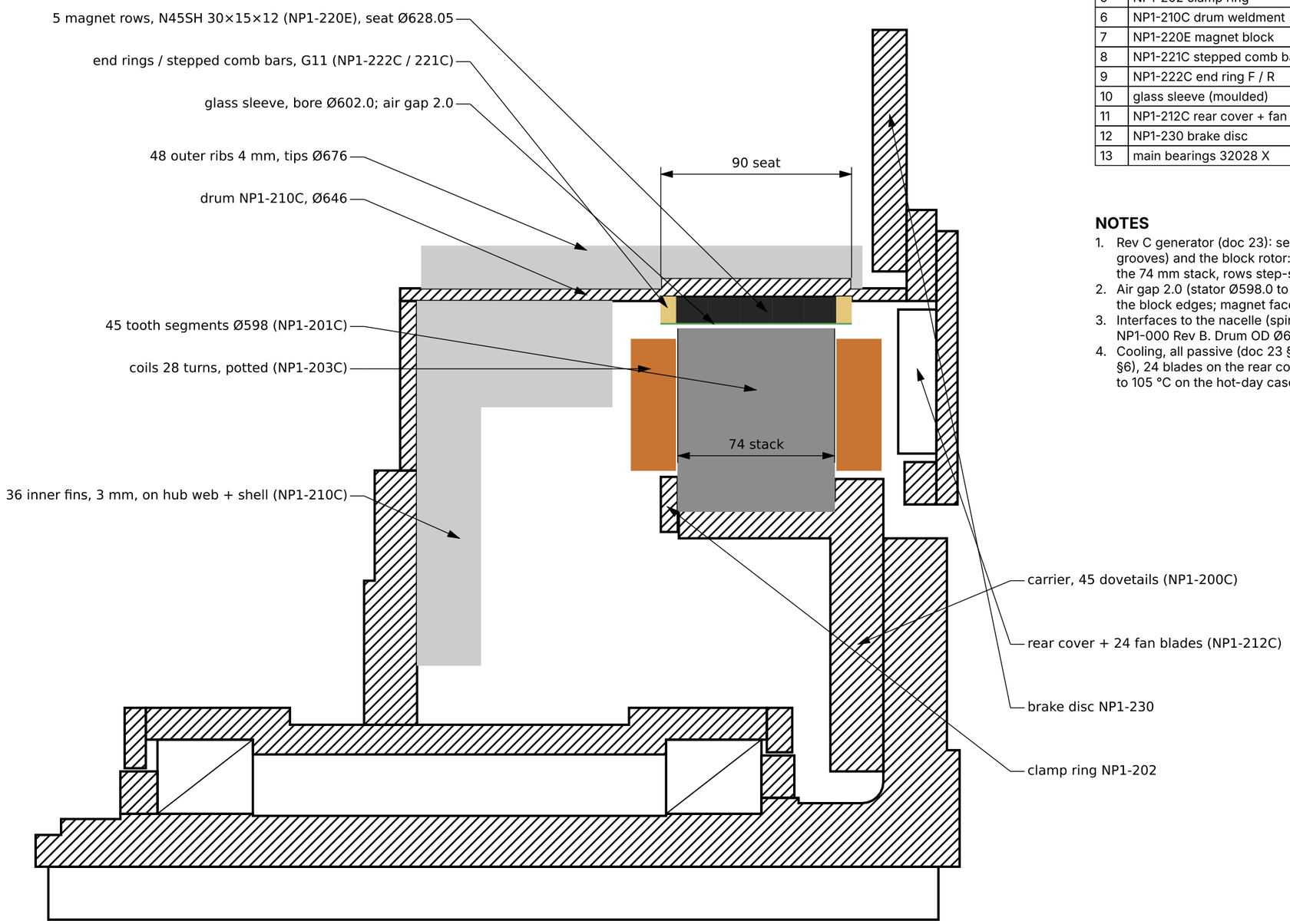
GENERATOR SECTION (upper half, 1:2.2)



#	Part	Qty	Material / note
1	NP1-100 spindle (stationary)	1	42CrMo4+QT
2	NP1-200C stator carrier	1	S355J2+N, 45 dovetails
3	NP1-201C tooth segment stack	45+3	M270-35A bonded, 205 sh.
4	NP1-203C coil on segment	45+3	28 t × 4 × Ø1.55, cl. 200
5	NP1-202 clamp ring	1	1.4404
6	NP1-210C drum weldment	1	S355J2+N, seat Ø628.05
7	NP1-220E magnet block	200+10	N45SH 30×15×12
8	NP1-221C stepped comb bar	40+4	G11
9	NP1-222C end ring F / R	1+1	G11
10	glass sleeve (moulded)	1	E-glass/epoxy 0.5–0.9
11	NP1-212C rear cover + fan	1	5083, 24 blades
12	NP1-230 brake disc	1	C45E+N
13	main bearings 32028 X	2	bought

NOTES

1. Rev C generator (doc 23): segmented stator of doc 18 (45 tooth segments in carrier dovetail grooves) and the block rotor: one N45SH block 30 × 15 × 12 per pole across, five rows along the 74 mm stack, rows step-skewed 0.20° by the stepped comb bars.
2. Air gap 2.0 (stator Ø598.0 to sleeve bore Ø602.0); sleeve 0.52 at the block centre, 0.89 at the block edges; magnet face r 301.52.
3. Interfaces to the nacelle (spindle, bearings, hub flange, disc, bedplate) unchanged from NP1-000 Rev B. Drum OD Ø646 unchanged.
4. Cooling, all passive (doc 23 §4 and §4a): carrier → spindle → finned insert path (doc 14 §6), 24 blades on the rear cover, 36 inner fins and 48 outer ribs on the drum; hot spot 94 to 105 °C on the hot-day cases (limit 130).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

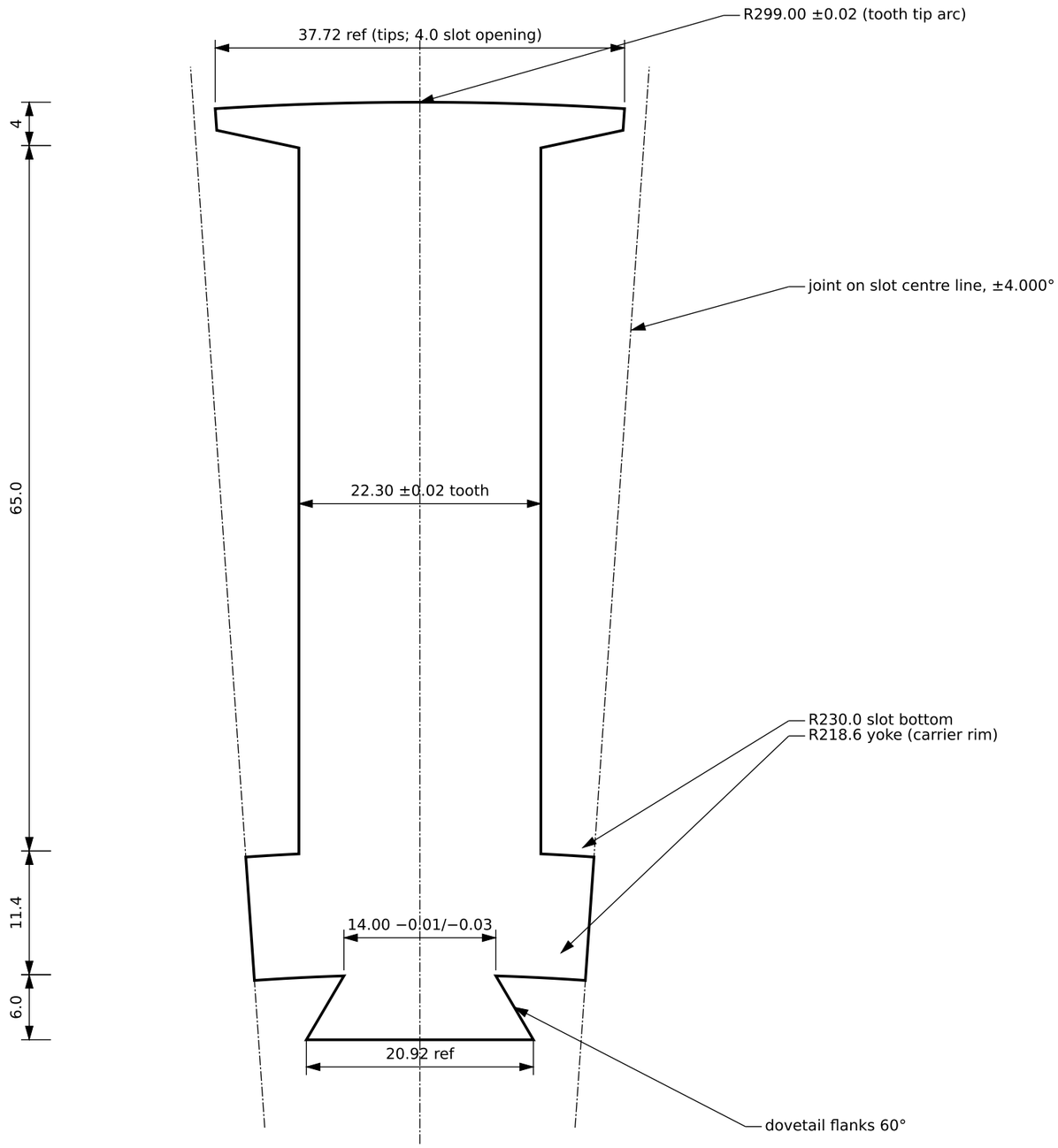
GENERATOR REV C · ASSEMBLY SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-G00	C · 2026-11-05	1:2.2
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see parts list	≈ 330 kg drivetrain (no hub)	Claude / not checked

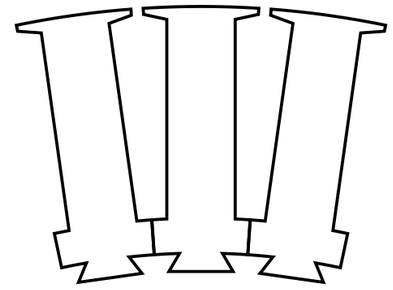
NOTES

1. DXF layer OUTLINE = cutting contour. Laser cut (P1), burr ≤ 0.02, heat zone ≤ 0.1; or progressive die (series).
2. Magnetic contour identical to the doc 14 one-piece lamination; joints on the slot centre lines in the yoke (8.000° pitch).
3. Stack: 205 sheets, 74.0 ±0.1 (sheets counted, height measured per segment), bonded in fixture NP1-T03 at 1.0 to 1.5 MPa, varnish cure per supplier.
4. Checks per stack: tip radius R299.00 ±0.02 on gauge, foot width 14.00 −0.01/−0.03, slot opening edges straight 0.05 over 74; Franklin test ≥ 10 Ω·cm².
5. Do not machine tooth tips after stacking. 45 + 3 spare segments per stator; all identical.

SEGMENT (2:1), radial direction up



3 OF 45 SEGMENTS (1:2)

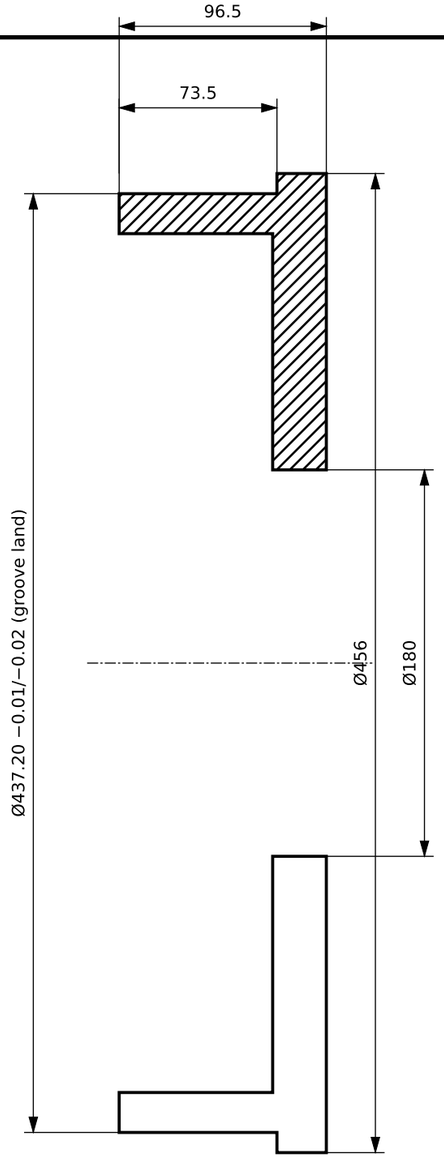


Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

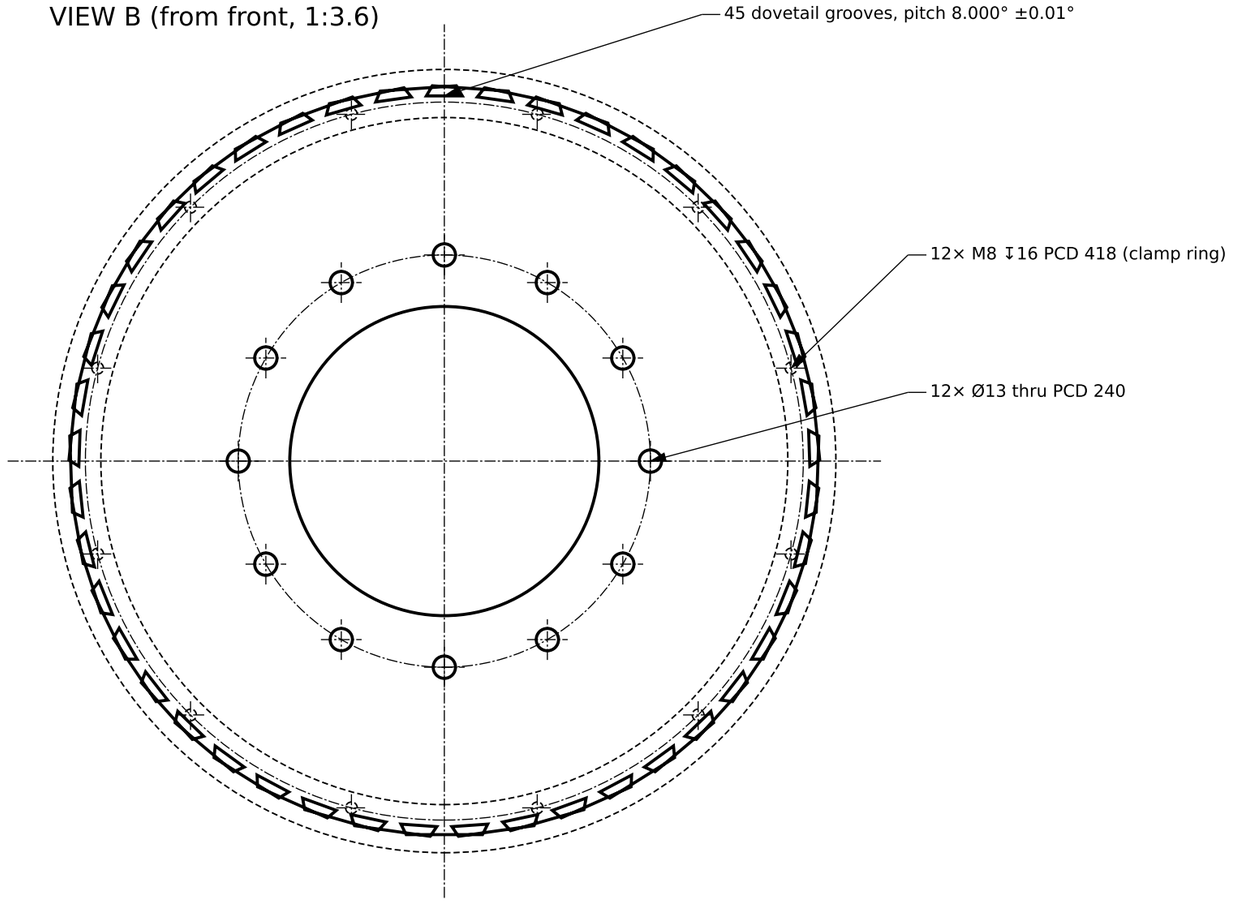
STATOR TOOTH SEGMENT LAMINATION (45 + 3 PER STATOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-201C	C · 2026-11-05	2:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
M270-35A, 0.35 mm, self-bonding varnish	1.07 kg per stack	Claude / not checked

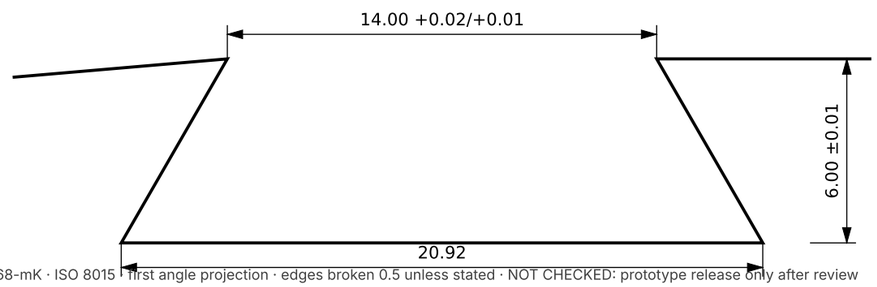


SECTION A-A (1:2.86)

VIEW B (from front, 1:3.6)



DETAIL C: GROOVE (5:1), flanks 60°



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

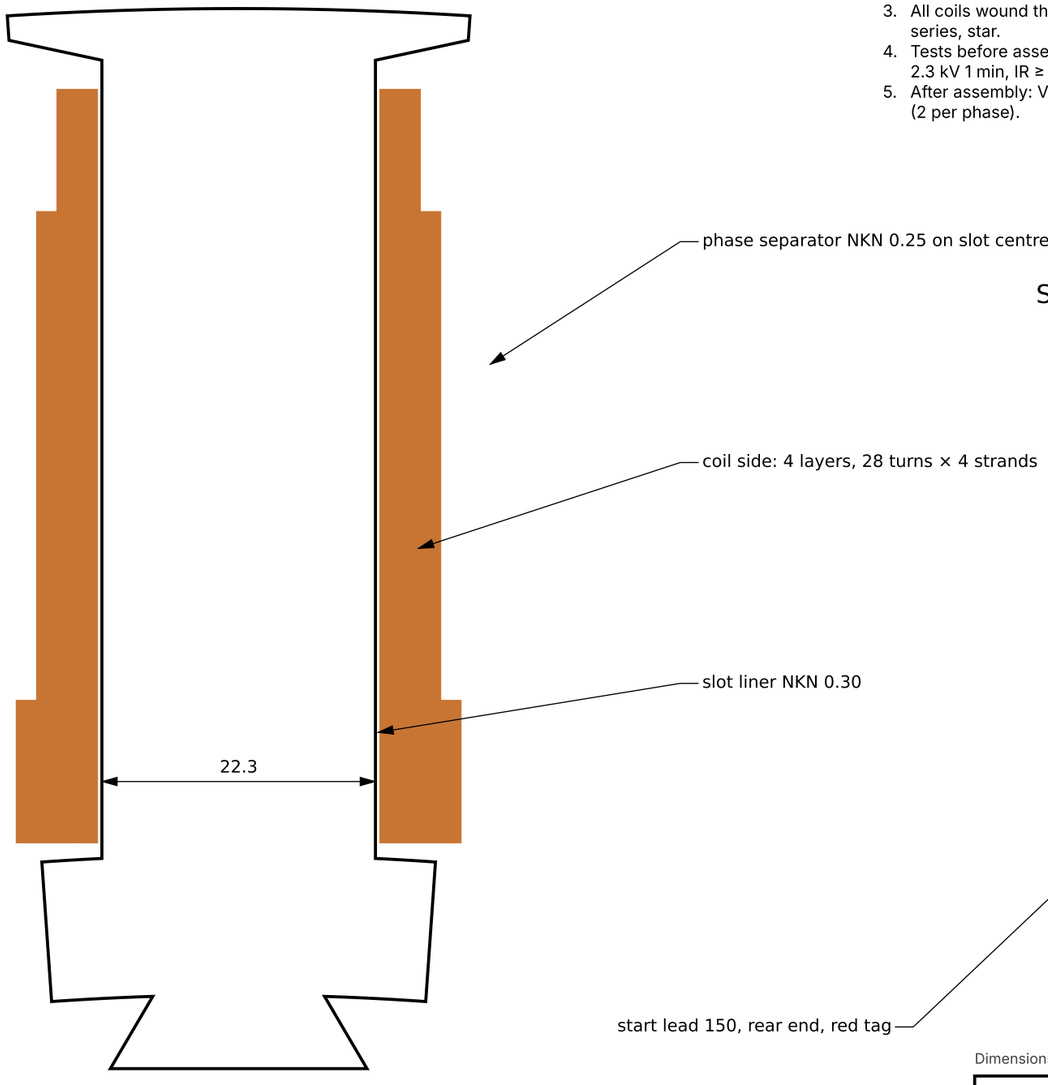
1. Rev C (doc 18 §5.1): grooves replace the shrink fit and the 3 keys. Grooves open to the front (x 343.5), closed by the rear stack stop (x 417); broached or wire-EDM.
2. Groove angular position ±0.01°, groove seat radius R212.60 ±0.01 to the bolting-face axis; foot clearance 0.01 to 0.03 (slides by hand when clean), fixed by VPI resin, clamped by NP1-202 (12× M8 A4-80, 20 Nm).
3. Run-out of groove lands and rear stop face to the bolting face and Ø180 spigot ≤ 0.02; bolting face flatness 0.02.
4. Bolt to the spindle flange: 12× M12 10.9 + hardened washers, 110 Nm, 2 fitted dowels Ø10 H7 PCD 240; Al-filled paste 0.05 on the full face (heat path, doc 14 §6).
5. Stress relieve 580 to 620 °C, 2 h, before final machining. No paint on grooves and bolting face; elsewhere epoxy primer 60 µm.

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR CARRIER WITH 45 DOVETAIL GROOVES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-200C	C · 2026-11-05	1:2.86 / 1:3.6 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N ring forging or plate, stress relieved	40.6 kg	Claude / not checked

COIL SIDES IN THE SLOT (2:1)

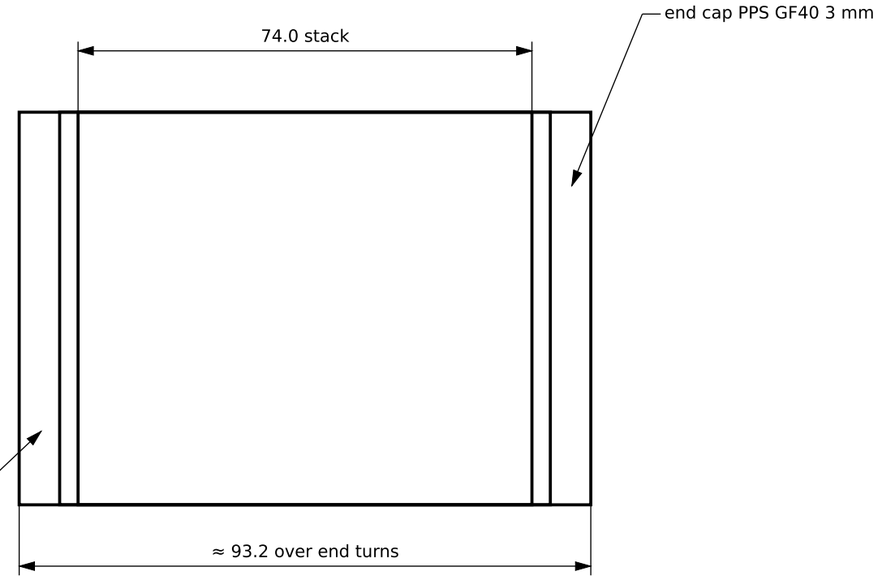


Layer (from tooth flank)	Positions	Used
1	37	37
2	37	37
3	31	31
4	17	7
5	3	0 (reserve)
Total (28 turns × 4 strands)	125	112

NOTES

1. Wind on chuck NP1-T04 by the dovetail, 4 parallel strands Ø1.55 (≤ 1.66 over enamel), 25 to 30 N per strand, layer by layer from the tooth flank, ≤ 60 rpm (doc 18 §5.6).
2. 28 turns per coil (Rev C keeps doc 18): 373 V line at 178 rpm, magnets 100 °C; 389 V at 60 °C. Wire per coil ≈ 6.6 m per strand; R20 per phase 0.217 Ω.
3. All coils wound the same direction; reversal at connection (doc 18 Appendix A): 15 coils per phase in series, star.
4. Tests before assembly (doc 18 §5.7): resistance ±2 % of the lot mean, surge 3.5 kV, hipot to the segment 2.3 kV 1 min, IR ≥ 1 GΩ.
5. After assembly: VPI class H epoxy, end windings potted (doc 18 §5.11, §5.12). PT100 in 6 separator gaps (2 per phase).

SIDE VIEW OF COIL ON TOOTH (1:1, simplified)



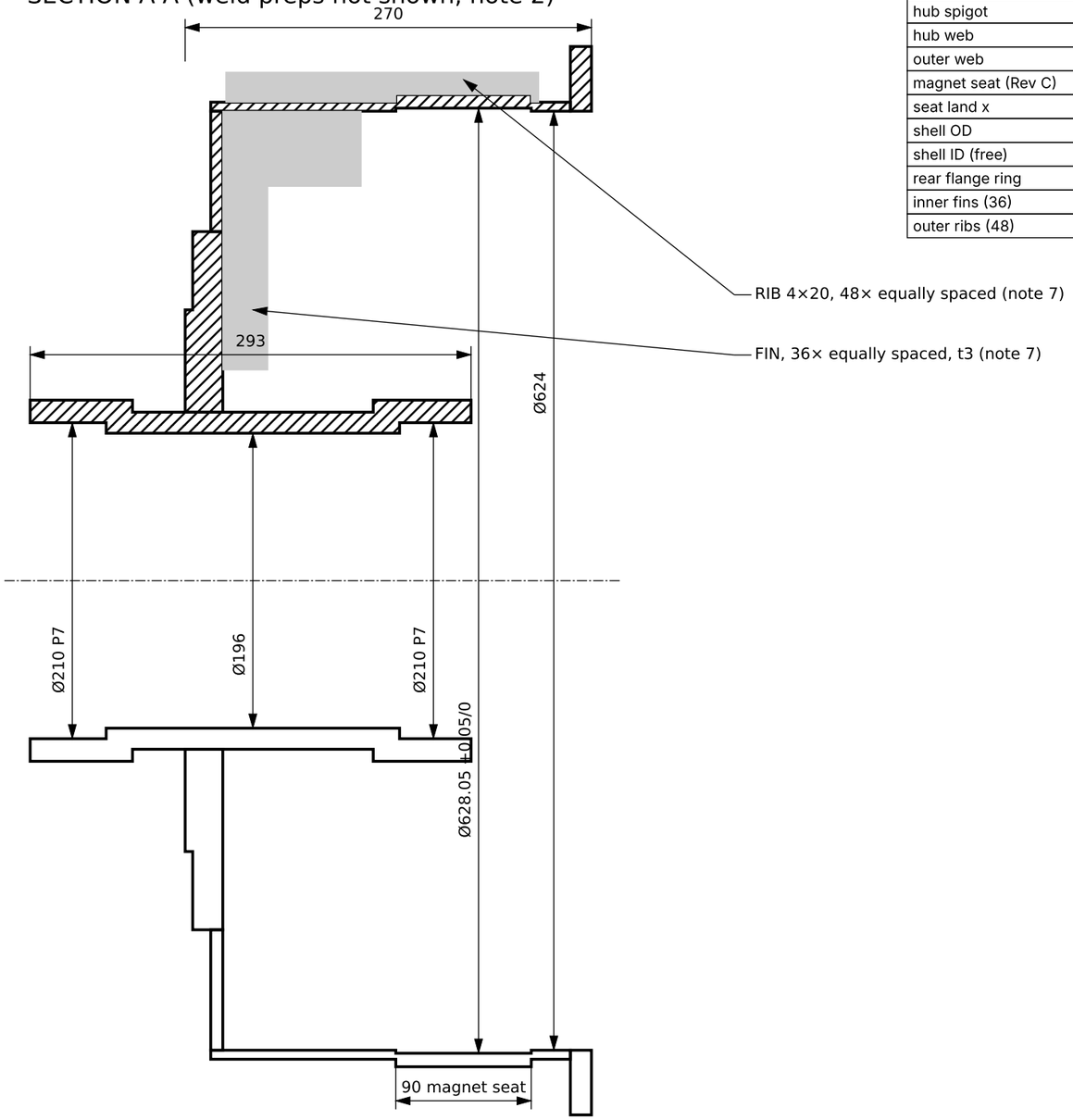
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

COIL ON SEGMENT (WOUND SEGMENT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-203C	C · 2026-11-05	2:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Cu enamelled Ø1.55 grade 2, class 200; NKN 0.30; PPS 4540 caps	≈ 0.45 kg Cu per coil	Claude / not checked

SECTION A-A (weld preps not shown, note 2)



Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat (Rev C)	Ø628.05 × 90	+0.05/0, coax. 0.02, Ra 3.2
seat land x	335 to 425	as Rev B
shell OD	Ø646	steel under seat 8.98
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø710 × 14	M10 PCD 644 + M10 PCD 678
inner fins (36)	t3, x 220-312, r 140-312	L-plate, web + bore, every 10°
outer ribs (48)	4 × 20, x 222-430	tips Ø676, every 7.5°

NOTES

1. Rev C: magnet seat Ø628.05 +0.05/0 (was Ø618.40) for the 12 mm N45SH blocks; steel under the seat 8.98 (FEA: 1.45 T median, 1.7 T at the pole edges, included in the EMF).
2. Welds full penetration at web/housing and web/shell, ISO 5817 B, 100 % MT, UT of the shell seam (ring-rolled shell preferred, doc 18 §4.2). Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up: cup seats Ø210 P7, magnet seat, hub spigot Ø360 f7, rear flange face; coaxiality ≤ 0.02, shoulders square ≤ 0.01.
4. Hub web 12× M16 thru PCD 400; rear flange ring 24× M10 PCD 644 (rear cover) and 24× M10 PCD 678 (disc).
5. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside zinc spray + epoxy (C5).
6. Balance finished rotor (drum, magnets, sleeve, disc, cover) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Passive cooling (doc 23 §4a): 36 inner fins t3 S355, L-shaped, fillet-welded both sides to hub web and shell bore, every 10°, 9 mm clear of the end windings; 48 outer ribs 4 × 20 S355 on the shell outside, every 7.5°, intermittent fillet welds 30/60 both sides, ends rounded. Weld fins and ribs before stress relief and machining.
8. Rear flange ring welded to the shell full round, continuous: with the hub web it keeps the 9 mm shell round against the magnet pull (doc 23 §5). Seat roundness 0.05 after stress relief.

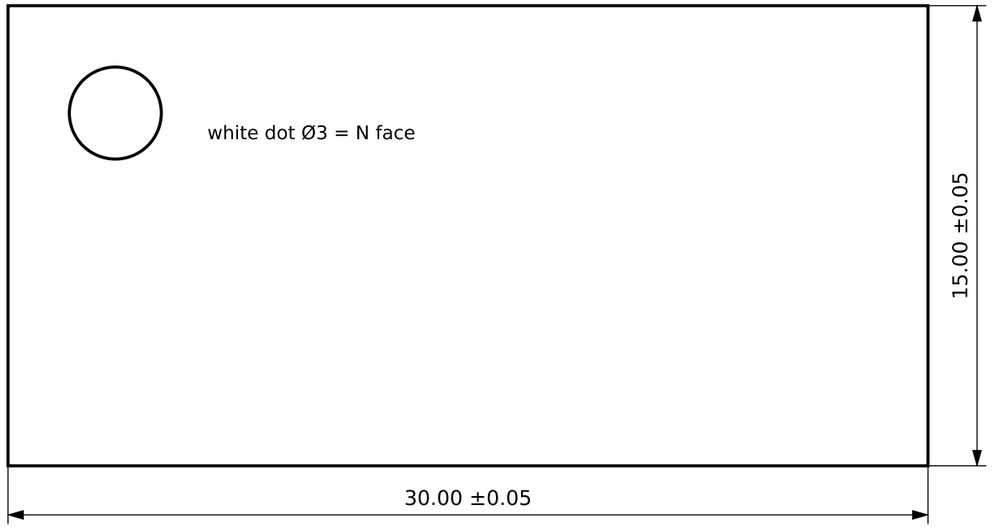
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

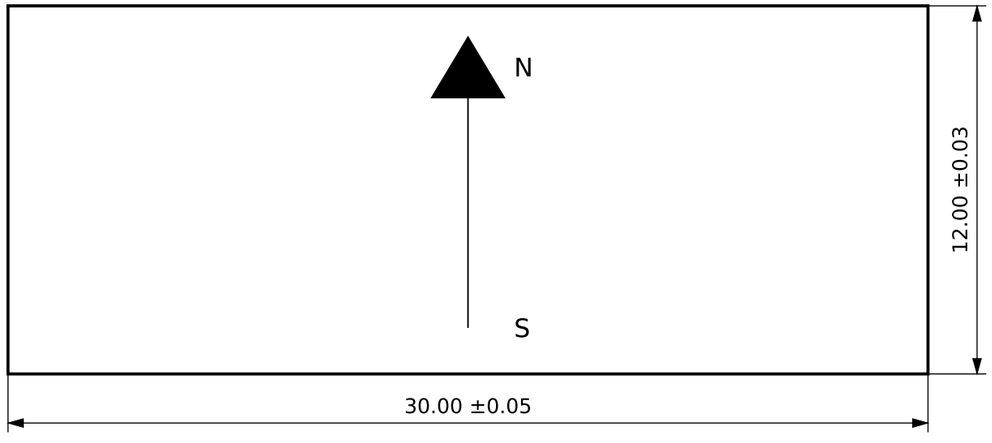
ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210C	C · 2026-11-05	1:3.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, welded, stress relieved	91.8 kg + 12.5 kg fins/ribs	Claude / not checked

VIEW A: N FACE (5:1)



VIEW B: END, magnetised through 12 (5:1)



Item	Requirement
Grade	N45SH: Br ≥ 1.32 T, HcJ ≥ 1,592 kA/m (20 kOe), (BH)max ≥ 342 kJ/m³ at 20 °C
Curves	J-H and B-H per batch at 20, 60, 100, 120 °C; αBr, βHcJ stated
Magnetisation	Through the 12 mm thickness, saturated; axis within 2° of the face normal
Tolerances	30 ±0.05, 15 ±0.05, 12 ±0.03; faces flat 0.02, parallel 0.02
Edges	Chamfer 0.2 to 0.3 × 45° all 12 edges; no chips > 0.3
Coating	NiCuNi 10 to 20 µm; 72 h neutral salt spray clean; adhesion OK with the epoxy
Uniformity	Moment of each piece within ±1.5 % of the lot mean; value per serial
Marking	White dot Ø3 on the N face; serial on the packaging position
Quantity	200 + 10 spare per generator; 8 extra for Q1 tests
Packing	Repelling layout in non-magnetic spacers; air freight per IATA PI 953

NOTES

1. One part for N and S poles: N poles face N towards the stator, S poles are the same block flipped (dot hidden against the seat).
2. Solid block (not two 6 mm pieces bonded). If a maker offers only 30 × 15 × 6, the doc 20 bonded pair is the fall-back with identical magnetics.
3. Five blocks per pole along the stack (15 mm pieces = axial segmentation for eddy loss, 4.7 W at rated).
4. Release of N45SH for production is conditional on the 100 °C design basis (doc 19 §16.3, doc 23 §4): bench magnet temperature ≤ 75 °C, temperature watch, no deliberate phase shorting.

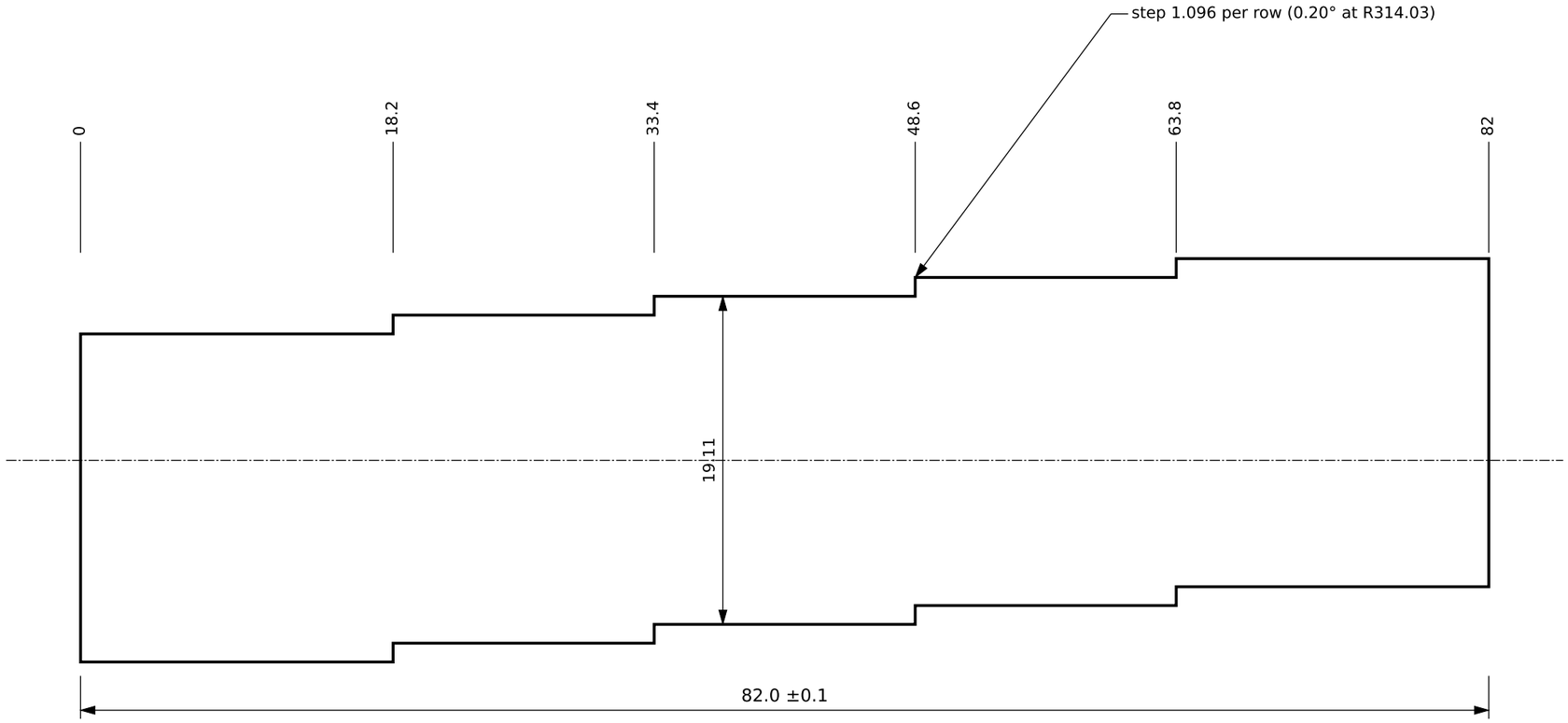
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

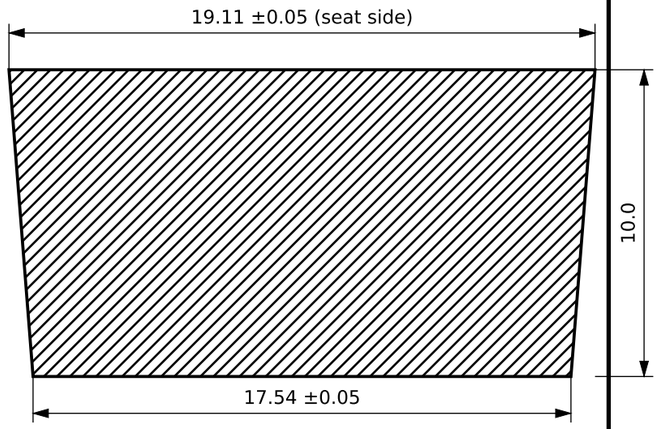
MAGNET BLOCK 30 × 15 × 12 (PURCHASE DRAWING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-220E	E · 2026-11-05	5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Sintered NdFeB N45SH, NiCuNi	41.0 g each, 8.21 kg per generator	Claude / not checked

PLAN, seen from the stator (3:1). Steps set the row skew.



SECTION (5:1), flanks follow the block sides



NOTES

1. One bar per pole gap (40 per rotor + 4 spare). Bar sits on the seat (wide face), bonded with the magnet epoxy; ends engage 4 mm pockets in the end rings NP1-222C.
2. Row boundaries (ordinates from the front end): steps of 0.20° each; the bar is identical for every gap. Mark the front end.
3. Side faces are the datum for the blocks: 0.10 glue clearance each side (comb width = gap between neighbouring blocks − 0.20).
4. G11 (class H), ±0.05 on widths; edges deburred, no delamination. Prove on the Q1 panel (doc 18 §10).

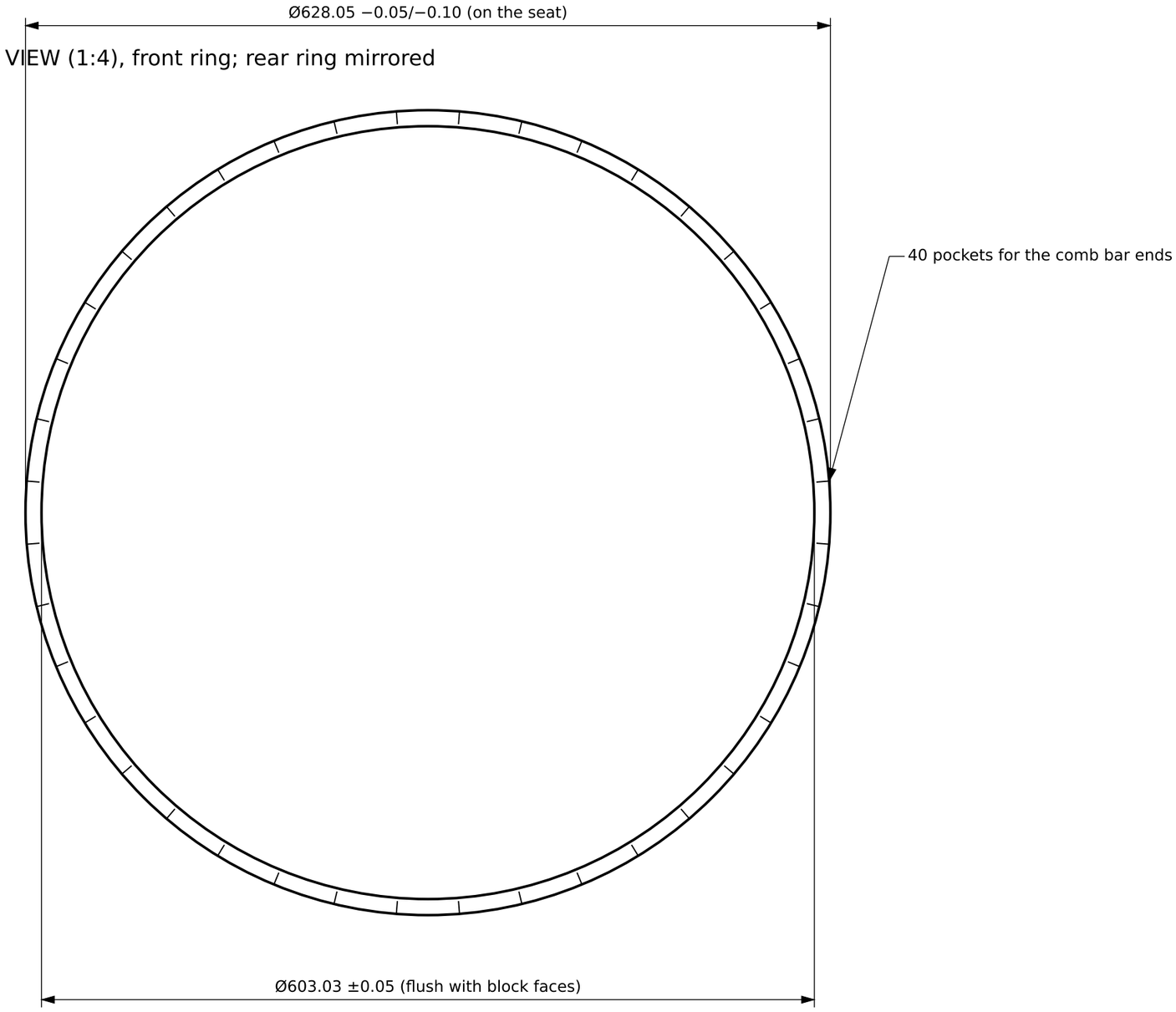
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

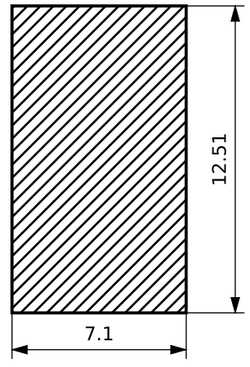
STEPPED POLE COMB BAR (40 + 4 PER ROTOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-221C	C · 2026-11-05	3:1 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
G11 glass-epoxy sheet, CNC machined	≈ 25 g	Claude / not checked

VIEW (1:4), front ring; rear ring mirrored



SECTION (4:1)



NOTES

1. Front ring at x 335 to 342.1, rear ring at x 417.9 to 425 (both 7.1 wide). They close the magnet pack and hold the comb bar ends.
2. Pockets: 19.11 wide (seat side), 10 deep radially, 4 deep axially, on the gap centre lines (4.5° + n × 9°), front ring rotated −0.40°, rear ring +0.40° (row skew).
3. May be made in 4 bonded arcs if no G11 tube of this size is available; joints on pocket centre lines.
4. Bonded to the seat with the magnet epoxy, then covered by the moulded glass sleeve.

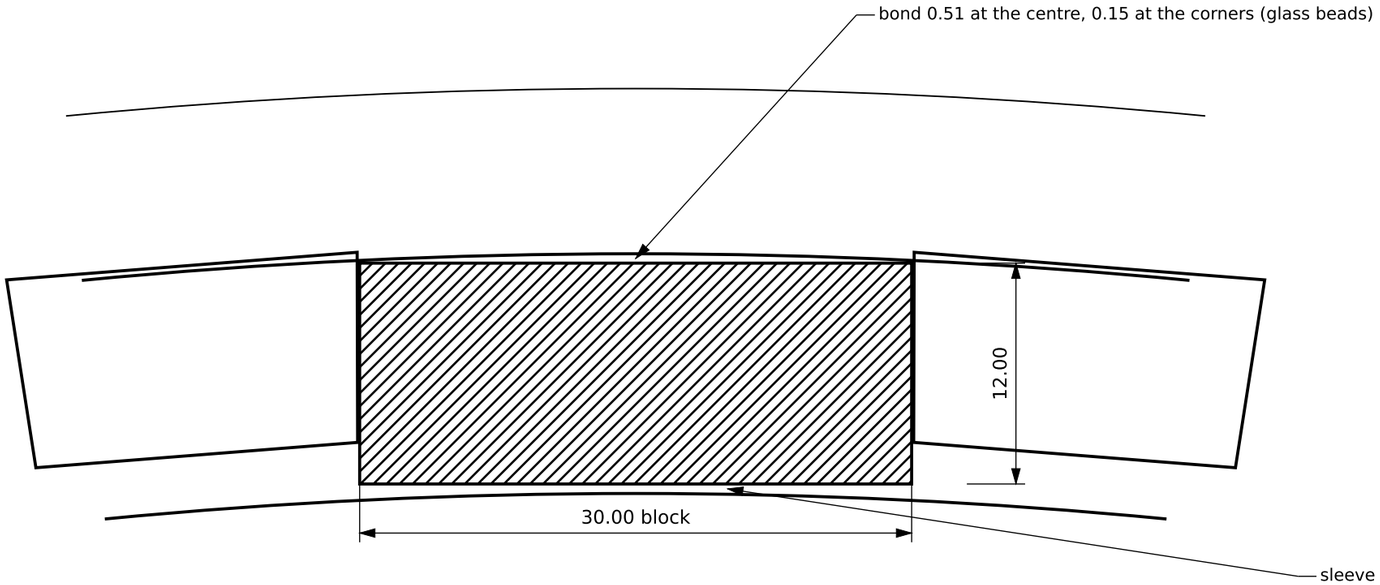
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

MAGNET END RINGS, FRONT AND REAR

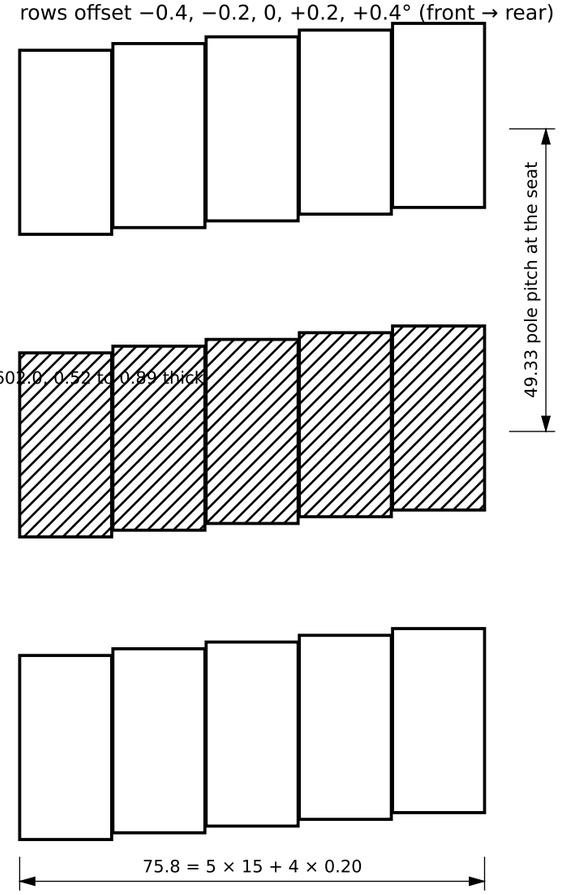
DRAWING NO.	REV / DATE	SCALE (A3)
NP1-222C	C · 2026-11-05	1:4 / 4:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
G11 glass-epoxy, machined from tube or segments	≈ 0.12 kg each	Claude / not checked

POLE SECTION (3:1), centre line = pole N1



LAYOUT, 3 POLES UNROLLED (1:1), front row left

rows offset −0.4, −0.2, 0, +0.2, +0.4° (front → rear)



NOTES

1. Bond: two-component toughened epoxy, RT cure, 80 °C post-cure, with 0.15 glass beads (doc 18 §4.2). Blocks placed with jig NP1-T02 against the comb bar faces; all N poles first, then S.
2. Sequence per pole: comb bars and end rings bonded and cured first; blocks row by row from the front; check each row against the template (skew 0.20 ±0.02° per row).
3. Pole map NP1-T10 after cure: every pole centre within ±0.05° of nominal, flux per pole within ±1.5 %.
4. Sleeve: E-glass/epoxy moulded on the Al mandrel NP1-T06 located on both cup seats; bore Ø602.0 ±0.03, thickness 0.52 at the block centres, 0.89 at the edges; covers the end rings.
5. Q1 check (doc 18 §10): sleeve thickness by section 0.45 to 0.95; air gap feeler check 2.0 ±0.1 all round after joining.

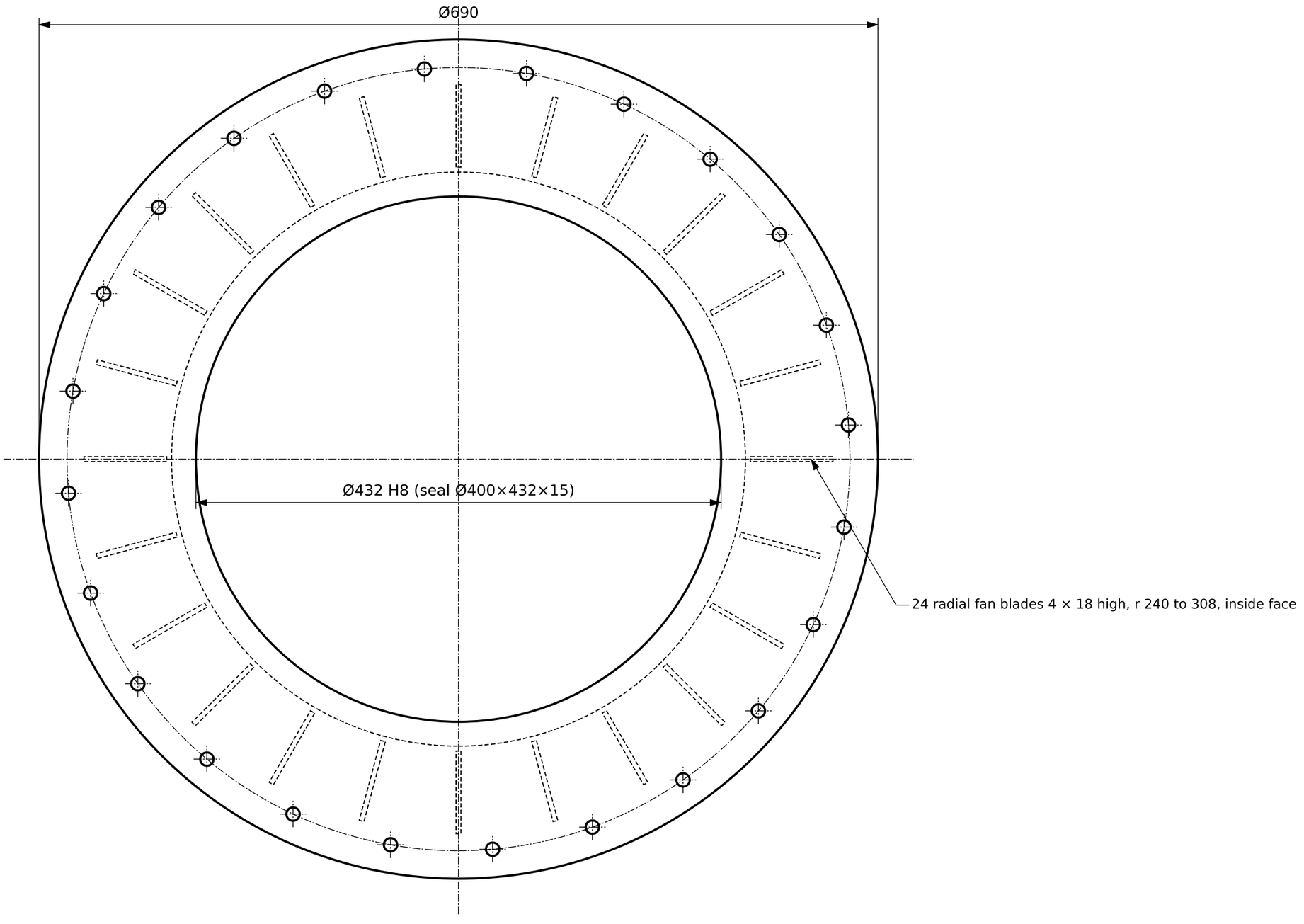
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

ROTOR MAGNET PACK: POLE SECTION, LAYOUT AND SLEEVE

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-223C	C · 2026-11-05	3:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
assembly	8.2 kg magnets	Claude / not checked

VIEW FROM FRONT (inside), plate 10, seal ring 15 × 20 at x 450 to 465



NOTES

1. Rev C (doc 23 §6): 24 radial blades welded (TIG) or bolted to the inside face; they stir the internal air over the end windings and carry heat to the cover (h_int ≈ 40 W/m²K instead of 15, to be measured on the bench, doc 15 T5).
2. Blades span x 447 to 465: 8 mm clear of the potted end turns (x ≤ 439) and of the carrier rear stop (x 440); tips r 308 run 4 mm inside the shell bore Ø624.
3. Rear seal: radial lip seal Ø400×432×15 HNBR −40 °C on the spindle flange OD; breather port M24×1.5 with desiccant breather.
4. Balance with the rotor (NP1-210C note 6). Inspection port Ø60 with cover at 180° (not shown).

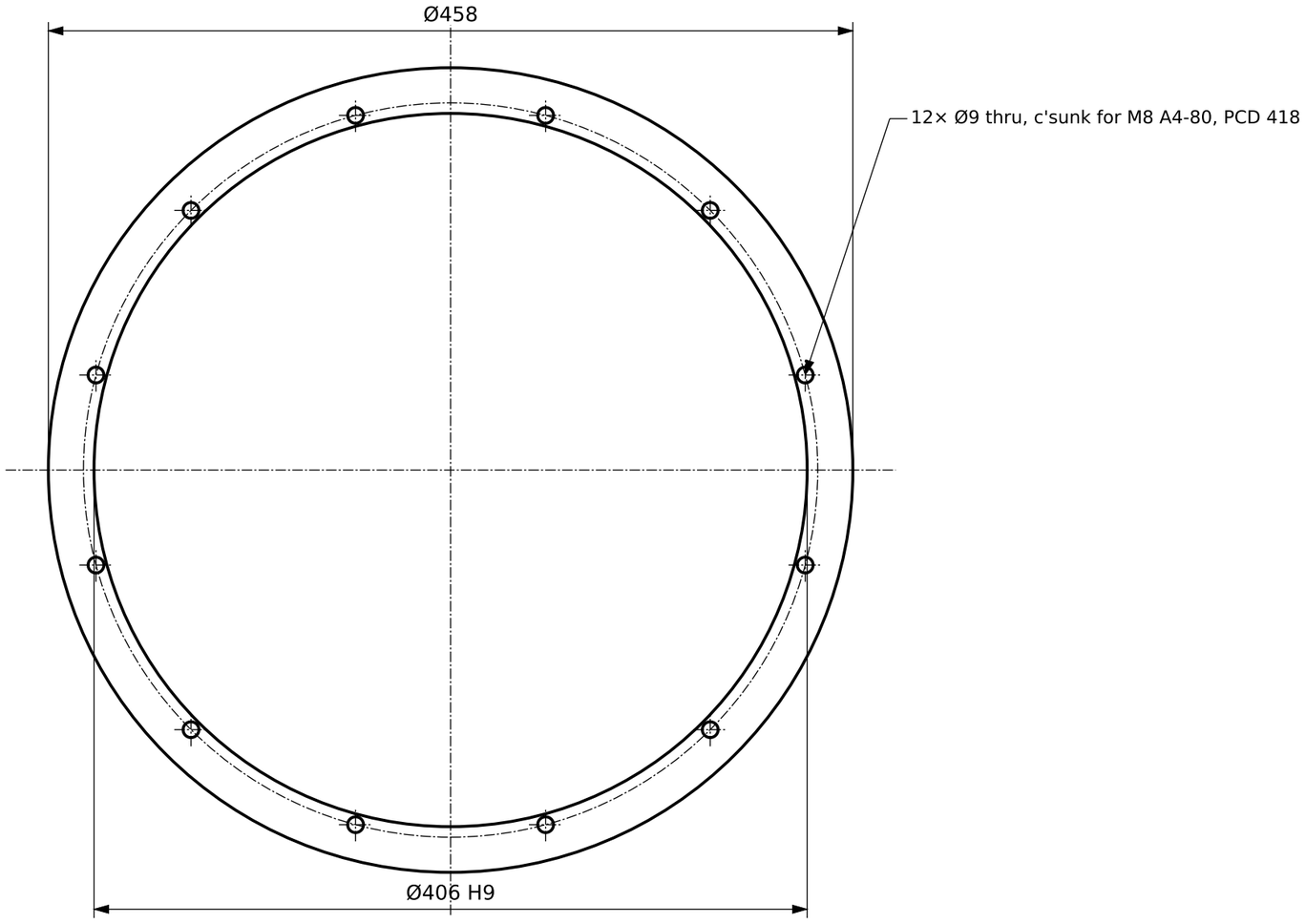
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

DRUM REAR COVER WITH SEAL CARRIER AND FAN BLADES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-212C	C · 2026-11-05	1:3.6
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
EN AW-5083-H111 plate, ring and blades	≈ 9.0 kg	Claude / not checked

VIEW (thickness 8.0, faces parallel 0.05)



NOTES

1. Laser cut from 8 mm plate, faces machined flat 0.05.
2. Must not touch the coils: inner end-winding radius ≥ 232; ring OD 229.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR FRONT CLAMP RING

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-202	B · 2026-11-05	1:3.3
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
1.4404 (non-magnetic)	2.2 kg	Claude / not checked